

A Decentralized Partially Observable Markov Decision Process for complete coverage onboard multiple shape changing reconfigurable robots

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ABSTRACT

Achieving complete area coverage by multiple collaborative robots is an essential aspect of productive deployment for applications such as cleaning, maintenance, and patrol. However, in real-world scenarios, physical constraints such as structural obstacles in the environment that generate narrow spaces, hinder the area coverage achieved by the fixed morphology robots, resulting in ineffective deployments. This paper proposes a Multi-Agent Reinforcement Learning (MARL) framework leveraging the Decentralized Partially Observable Markov Decision Process (Dec-POMDP) for Complete Coverage Planning (CPP) of multiple shape-reconfigurable robots. This decentralized framework runs onboard each robot to mitigate the physical limitations while achieving complete area coverage. The proposed approach was trained on diverse map environments with different tight spaces to generate policies that enabled the robots to learn and coordinate their joint actions. The framework was further demonstrated on multiple unseen test environments to evaluate the generalization capabilities and area coverage performance. Additionally, baseline comparisons with the different combinations of multiple fixed morphology robots were conducted to validate the area coverage effectiveness of the proposed approach. In all scenarios, the proposed framework achieved 100% area coverage while the baselines only achieved partial area coverage ranging from 46.21% to 96.55%. The improved coverage performance compared to the baselines demonstrates the potential and usefulness of such an approach in the deployment of multiple reconfigurable robots.

1. Introduction

Robots have been readily deployed in different environments for a multitude of daily applications such as cleaning (Wijegunawardana et al., 2023), patrolling (Yeo, Pey, & Elara, 2023), auditing (Pey, Povendhan, Pathmakumar, & Elara, 2022), and maintenance (Ezhilarasu, Pey, Muthugala, Budig, & Elara, 2024; Wang, Liu, Liu, Ling, & Zhang, 2023), to supplement humans in these tasks. Within robotics, the Complete Coverage Planning (CCP) problem is still a persistent challenge that requires a robot to cover all points in a state workspace without obstacle collision (Galceran & Carreras, 2013; Gao, Lv, Yan, & Han, 2022). Achieving CCP is essential for the above-mentioned applications as the robots should access all possible areas including complex or unknown environments for accomplishing task requirements (Tan, Mohd-Mokhtar, & Arshad, 2021).

In path planning, CCP is achieved through the usage of primarily classical, metaheuristic, or artificial intelligence algorithms (Lin, Liu, Wang, & Kong, 2022; Liu et al., 2023; Tan et al., 2021). For example, in classical algorithms, Jan, Luo, Hung, and Shih (2014) and Choset

(2000) demonstrated cell decomposition approaches, such as vertical cell decomposition with convex hull and boustrophedon cellular decomposition to segment the map into sections. The approach was then supplemented by either spiral (Stachniss, 2009) or zigzag (Liu, Lin, & Zhu, 2008) motions to allow complete coverage in a sequential manner. Abdallaoui, Aglzim, Chaibet, and Kribèche (2022) and Chen, Luo, Liu, Zhao, and Liang (2021) explored sampling-based methods such as Rapidly-exploring Random Tree (RRT) and Probabilistic Road Map (PRM) respectively for high dimension spatial areas, but struggle with planning efficiency. Metaheuristics such as Genetic Algorithm (GA) (Goldberg, 2013), Particle Swarm Algorithm (Gad, 2022) (PSO), and Grey Wolf Optimization (GWO) (Mirjalili, Mirjalili, & Lewis, 2014) are often utilized to solve CCP where optimization is required. For example, to achieve CCP, Wu et al. (2023) implemented a GA-based approach for self-driving agricultural robots using a multi-objective equilibrium function while Wu, Wang, and Guo (2024) implemented a stretched fitness function, an adaptive mutation operator, and a crossover operator for an unmanned surface vehicle. In contrast, Melo,

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Junior, and Oliveira (2018) proposed exploration-enhanced PSO algorithms to coordinate collaborative multi-target searching. Kamalova, Navruzov, Qian, and Lee (2019) similarly documented integrated GWO approaches for searching regions of uncertainty for collision-free robot path planning. However, the approaches only achieved partial area coverage and were limited to search-based exploration or point-to-point navigation tasks. Lastly, artificial intelligence algorithms focus on approaches such as Glassius Bio-inspired Neural Network (Yao & Zhao, 2021), Reinforcement Learning (Lei, Luo, Jan, & Bi, 2022) and Fuzzy Logic (Caggiano, 2019) for online and offline coverage.

Multi-Robot Systems (MRS) have gained popularity in the robotics community due to the increase in effectiveness, efficiency, flexibility, and fault tolerance over single robot systems (Almadhoun, Taha, Seneviratne, & Zweiri, 2019; Antonyshyn, Silveira, Givigi, & Marshall, 2023; Roldán-Gómez & Barrientos, 2021). To enable increased area coverage and faster coverage time, multi-robot collaborative exploration was implemented by Lin and Huang (2021). Li et al. (2022) demonstrated task allocation for multiple robots with different velocities using a credit-based framework while Luo, Yang, Li, and Meng (2017) focused on a neural-dynamics approach for collaborative navigation.

With the recent successes and importance of Reinforcement Learning (Zhang & Mo, 2021) and Multi-Agent Systems (Leibo, Zambaldi, Lanctot, Marecki, & Graepel, 2017; Wang, Wan, Zhang, Li, & Zhang, 2016), Multi-Agent Reinforcement Learning (MARL) (Nguyen, Nguyen, & Nahavandi, 2020) is a rapidly expanding domain within robotics that combines these two aspects to enable group of robots to solve challenging tasks in high dimensional environments. Some applications of MARL in robotics include path-planning (Çetinkaya, 2021; Panov, Yakovlev, & Suvorov, 2018), manipulation (Nguyen & La, 2019), and locomotion learning (Haarnoja, Zhou, Abbeel, & Levine, 2018). MARL can also allow better potential outcomes for all agents than a single agent could achieve on its own. Within MARL, agents have the capacity to make decisions and learn from experience to maximize their individual goals through interaction with other agents and the environment (Oroojlooy & Hajinezhad, 2023). MARL algorithms have the task of enabling each agent to learn an optimized policy, resulting in an action at each timestep, to achieve a common goal of the system (Busoniu, Babuska, & De Schutter, 2008). Unlike single-agent RL, MARL allows agents to communicate to exchange information to enhance learning if the assigned tasks are similar. Additionally, even if one agent fails within the environment, other agents can continue to operate to complete the goal. MARL is usually modeled through a few major properties (Orr & Dutta, 2023): (1) Type of control, (2) Type of observability, and (3) Agent behavior. The control type can be categorized into either centralized or decentralized. In a centralized situation, a common controller dictates the action of each agent. In contrast, a decentralized controller enables individual agents to make their own decisions. For observability, a partial observation limits only local information to each agent while a full observation allows global information to be inputted for each agent. Lastly, cooperative agent behaviors allow all agents to work together to achieve a common goal. However, for competitive scenarios, each agent chooses decisions to maximize its own benefit and goals while outcompeting other agents.

Most of the above-mentioned related work is conducted using fixed morphology robots. These robots are limited by the inability to maneuver through tight and narrow spaces, which could be caused by either obstacles or the physical structure of the environment (Muthugala, Samarakoon, & Elara, 2022, 2023). These factors impose low height and low width constraints required by the robots to navigate safely through these spaces. Additionally, certain narrow spaces may potentially be entry points to another section of the environment in which the robots are completely unable to explore that may contain further important information about the environment. Hence, given that only partial coverage is achieved from the environment, the robot deployment would be ineffective and unproductive. Furthermore, CCP would

still be ineffective if the specifications of fixed morphology robots do not have the capacity to change their morphology in order to navigate through tight spaces safely (Samarakoon, Muthugala, & Elara, 2022a). As such, reconfigurable robots mitigate the physical constraints through the unique capabilities of modularity or shape-changing to enable flexible alteration of their morphology (Samarakoon, Muthugala, & Elara, 2022b). Therefore, reconfigurable robots have been introduced to mitigate the limitations of fixed shape robots (Yim et al., 2007). While MARL has been actively explored within robotics, the existing research is primarily limited to fixed morphology robots. Additionally, there has been little or no research work on integrating MARL or demonstrating the usage of multiple reconfigurable robots for achieving CCP within the domain of reconfigurable robots.

By taking the considerations discussed above, the main methodologies and contributions of this paper are summarized as follows:

1. A first-of-its-kind work of CCP using multiple reconfigurable robots.
2. A novel MARL framework leveraging the Decentralized Partially Observable Markov Decision Process (Dec-POMDP) for CCP using multiple shape-changing reconfigurable robots. The framework enables robots to coordinate their joint actions by learning interactions with the environment.
3. Training policies of the proposed MARL framework in simulation and deployment of policies in unseen simulation environments.
4. Comparison of area coverage performance of the trained policy with fixed morphology robots baselines.

The structure of the paper is divided in 4 sections. Section 2 details the proposed MARL framework formulation for CCP using multiple intra-reconfigurable shape-changing robots called Smorphi. Section 3 documents the results from the training process and discusses the evaluation of the framework on new and unseen environments, as well as the comparison of performance with baselines. Section 4 concludes the paper and discusses the future works and scopes of the research.

2. Problem formulation

In this paper, we formulate the problem of complete area coverage using multiple reconfigurable robots as a Multi-Agent Reinforcement Learning (MARL) scenario. Shape-changing reconfigurable robots called Smorphi are used to ensure that the robots have the capacity to navigate through an environment consisting of tight spaces of different sizes.

2.1. Smorphi: A shape-changing reconfigurable robot

Derived through transformation design principles documented by Singh et al. (2009), the design of a Smorphi robot is inspired by guidelines and facilitators that enable mechanical transformation for reconfiguration behavior. Each Smorphi robot,¹ with the major components highlighted in Fig. 1(a), contains 4 modules, where a primary module (Module 2) controls and drives the locomotion of the secondary modules (Module 1, 3 and 4). The usage of mecanum wheels further enables the holonomic motion of each module during translation and shape changing reconfiguration processes. The presence of 3 hinges placed in fixed positions between adjacent modules, displayed in Fig. 1(b), allows for the rotation of modules to change the robot into a desired shape. During shape change, the solenoid on each module enables the attachment and detachment of adjacent units through a lock mechanism.

With these features, each Smorphi has the capacity to change between O and I shapes, shown in Fig. 2. The symmetrical nature of both shapes help to reduce the complexity during navigation and

¹ Product website: <https://wefaarobotics.com.sg/>.

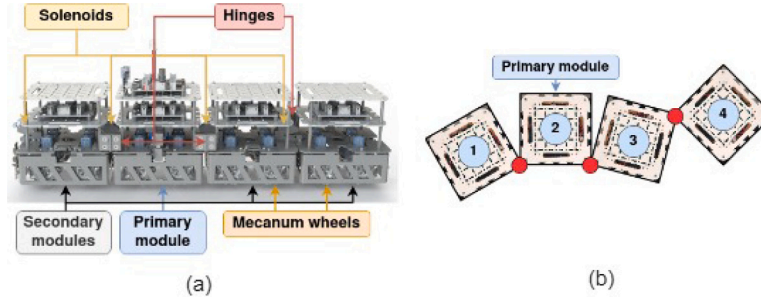


Fig. 1. (a) Single Smorphi robot with the highlighted major components across the 4 modules. (b) Top view display of the motion of each module about the 3 hinges (red dots) to enable shape changing.

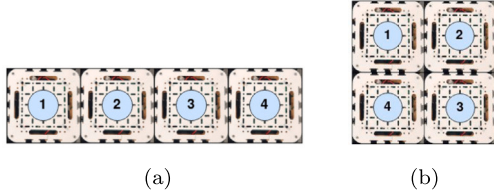


Fig. 2. (a) Smorphi robot in I shape configuration. (b) Smorphi robot in O shape configuration. Each module's position is labeled for each configuration.

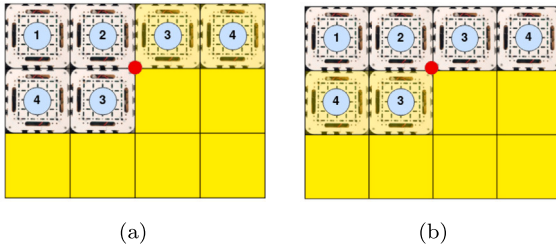


Fig. 3. (a) Workspace required for O to I shape change reconfiguration. (b) Workspace required for I to O shape change reconfiguration. Yellow spaces indicate the area required for the respective shape changes and translucent modules represent the final positions after shape change about the hinge (red dot).

coverage deployments. However, to ensure successful shape change, the robot must first check for a sufficiently big workspace that is available within the vicinity of the robot, displayed in Fig. 3, given the robot's current shape. The free workspace helps the robot to avoid hitting environmental obstacles or other robots nearby. During shape change, modules 3 and 4 are rotated 180° about the hinge while modules 1 and 2 retain their positions. The shape changing process provides the flexibility of enabling the robot to navigate through different sizes of narrow spaces.

2.2. Decentralized partially observable Markov decision process

Within robotics, the objective of MARL is to allow a group of autonomous agents to interact with the environment and learn behaviors to achieve a set of goals. In single-agent representations, a Markov Decision Process (MDP) is used to formulate the environment given that the rewards and state transitions are fixed, which satisfies the stationarity assumption. In contrast, in a multi-agent setting, all agents interact with and learn from the environment concurrently, this results in non-stationary state dynamics and expected rewards. The new state of the environment is therefore dependent on the joint actions of all agents beyond just a single agent's action and all agents must adapt their behavior to the other agents' actions and changing policies. This violates the stationarity assumption of the MDP and the representation of multi-agent environments using MDP would need to be extended

further.

An approach to select a suitable MDP variant for multi-agent settings would be to consider the nature of the interactions between all agents and the type of actions taken. Given that the goal of the robots is to achieve complete coverage, collaborative behavior is encouraged among the robots to maximize the joint cumulative rewards. In an actual deployment, each robot is vulnerable to limited sensing and perception due to the limitations of the hardware sensors onboard. Hence, it would not be practical for the robots to view the full ground truth spatial area. Instead, a partial observation of their direct vicinity and range can be used to provide the robots with information of their direct surroundings. Considering these major aspects of the environment formulation, the Decentralized Partially Observable Markov Decision Process (Dec-POMDP) will be used to model the MARL CCP problem.

Other models include Team Markov Games (TMG), Partially Observable Markov Games (POMG), and Extensive Form Games (EFG). TMG involves multiple agents grouped into groups for cooperative or competitive scenarios, however, the model assumes and requires a full observation of the environment state at each timestep. POMG is an extension of the TMG framework where agents operate in partially observable environments. However, unlike Dec-POMDP, the agents are competitive and each agent aims to maximize its own reward function, leading to conflicting goals between agents. EFG models interactions in a sequential manner, where each agent takes turns making decisions based on the game's history and imperfect information. This model focuses on a competitive setting using strategic interactions and assumes that agents can observe the history of the game. With these limitations, given that the context of this study requires collaborative and partially observable agents, Dec-POMDP is the most appropriate among the models.

Dec-POMDP is a fundamental model used in reinforcement learning for enabling coordination and decision-making among multiple decentralized agents. To model a Dec-POMDP, a tuple consisting of 8 parameters is conventionally used as discussed by Wong, Bäck, Kononova, and Plaat (2023) and Amato, Chowdhary, Geramifard, Üre, and Kochenderfer (2013). In this paper, we model the complete coverage using multiple reconfigurable robots based on these conventions. An overview of the Dec-POMDP contextualized for the complete area coverage problem is displayed in Fig. 4. At each timestep, every agent will interact with the global environment to update its observations. The observations are fed into a reinforcement learning algorithm to train a local policy to generate an action with the highest probability for the agent. The actions from all agents are inputted into a joint policy, which emerges from the information sharing across all agents and how the agents collectively behave based on the interactions and dependencies with one another. The actions from each agent are inputted into the joint policy simultaneously at each timestep to facilitate coordinated decision-making at the current state. The joint rewards are then calculated and returned as input to each agent's local policy. Using the joint actions, the state of the global environment is updated for

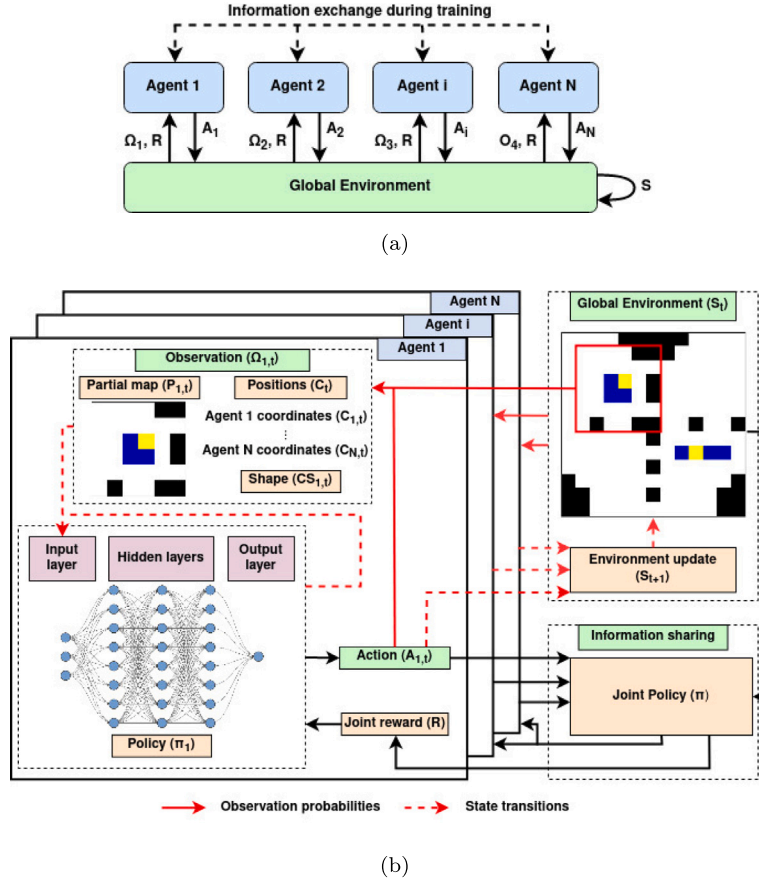


Fig. 4. Proposed complete area coverage framework using a Decentralized Partially Observable Markov Decision Process approach for reconfigurable robots: (a) Simplified framework diagram. (b) Detailed framework overview.

the next timestep. The robots also engage in parameter sharing of the network variables with one another to facilitate cooperative behavior during the training process. A centralized critic helps to evaluate the robots' global joint states and actions to guide and provide gradients to the local policies for updates across each timestep. The centralized training and decentralized execution approach helps to stabilize each agent's learning during training even though the policies of other agents are changing.

To model a Dec-POMDP, the parameter tuple consisting of $(I, S, A_i, T, R, \Omega_i, O, \gamma)$ are detailed as follows:

- I represents the finite set of N agents.
- S represents the finite state space for all agents with an initial state distribution of b_0 .
- A_i represents the finite set of actions for each agent, i .
- A represents the set of joint actions, where $A = \times_i A_i$ and $\times$ is the Cartesian product operator.
- T represents the state transition probability function, where $T: S \times A \times S \rightarrow [0, 1]$. This indicates the probability of transitioning from state $s \in S$ to $s' \in S'$ given that the set of actions of $\vec{a} \in A$ is executed by the agents. T is not explicitly defined but is calculated implicitly and modeled by environmental state transitions.
- R represents the given reward function, where $R: S \times A \rightarrow \mathbb{R}$. R is the immediate joint reward obtained for being in the state $s \in S$ using the set of actions $\vec{a}$.
- Ω_i represents a finite set of observations for each agent, i . Ω is the set of joint observations, where $\Omega = \times_i \Omega_i$.
- O represents the observation probability function, where $O: \Omega \times A \times S \rightarrow [0, 1]$. O is the probability of visualizing the set of observations of $\vec{o} \in \Omega$ if the set of actions of $\vec{a} \in A$ is taken

to obtain $s' \in S$. O is not explicitly defined but is calculated implicitly and modeled by updated observations and the agent's actions.

- γ represents the discount factor, which in the finite-horizon case, is usually equals to 1.

In a Dec-POMDP, each Smorphi robot is considered a unique agent in the global environment and has its own respective coordinates. Given that each Smorphi is capable of holonomic motion, the robot is able to perform translation upwards, downwards, left, and right. Additionally, the shape-changing ability of the robot is also added to the action set, shown in Eq. (1). n represents the iterator variable indicative of the 5 discrete actions that a Smorphi robot can perform.

$$A_i = \{A_{i,1}, A_{i,2}, A_{i,3}, A_{i,4}, A_{i,5}\}; \text{ where } A_{i,n} \in \{0, 1\} \quad (1)$$

To formulate the observation space, Ω_i , for each robot, 3 major parameters are considered. The first is the partial map, P_i , that represents the direct partial observation of the local area around the robot at the current state of the global environment at timestep t . The second parameter is the set of 2D coordinates, C , representing the positions of all robots. The position of a single robot, C_i , is represented as a 1×8 array comprised of the 2D coordinates of all 4 modules. To facilitate communication and coordination, each robot will have full access to the positions of other robots. The third parameter is the current shape of the robot, CS_i , to indicate to the robot to toggle between shapes to access tight spaces.

The rewards for each robot, R_i , upon interaction with the environment can be split into dense, $R_{d,i}$ and sparse $R_{s,i}$ reward categories as discussed by Mohtasib, Neumann, and Cuayáhuil (2021), which is displayed in Eq. (2). In the dense rewards category, the rewards are

given more frequently based on immediate steps or progress. In contrast, sparse rewards are only awarded based on the robots' collective progress when achieving a specific milestone or outcome to guide the robots generally towards achieving the goal of complete coverage.

$$R_i(S, A, S') = R_{d,i} + R_{s,i} \quad (2)$$

Under the dense rewards category displayed in Eq. (3), the agent is rewarded upon traversing an unvisited cell. Narrow spaces are identified as bonus cells and are given higher rewards as these areas require the robot to toggle between different shapes which promotes reconfiguration. In contrast, the robot is penalized when a cell is revisited or collides with an obstacle, the boundaries, or another agent.

$$R_{d,i} = \begin{cases} 10 ; & \text{if unvisited cell (per module)} \\ 30 ; & \text{if unvisited bonus cell (per module)} \\ -0.5 ; & \text{if visited cell (per module)} \\ -2 ; & \text{if collision (per agent)} \end{cases} \quad (3)$$

Under the sparse rewards category documented in Eq. (4), the agent is rewarded based on the total percentage of area covered, PAC . As the percentage covered progresses closer to complete coverage, rewards are given more frequently to incentivize the agent to find the unvisited cells, and a bonus reward is also given for achieving full coverage. To further encourage the robot to take a minimal number of steps, rewards are also given and scaled based on the total number of timesteps, t_s , taken to achieve complete coverage. A maximum timestep limit, t' , is imposed on t_s to prevent the agent from training in an endless recursive loop.

$$R_{s,i} = \begin{cases} 40 ; & \text{if } PAC = 25, 50, 60, 70, 80, 90 \\ 80 ; & \text{if } PAC = 95, 96, 97, 98, 99 \\ 200 ; & \text{if } PAC = 100 \\ 200 ; & \text{if } t_s < 200 \\ 150 ; & \text{if } t_s \geq 200 \text{ and } < 400 \\ 100 ; & \text{if } t_s \geq 400 \text{ and } < 600 \\ 50 ; & \text{if } t_s \geq 600 \text{ and } < 800 \\ 0 ; & \text{if } t_s \geq 800 \\ -200 ; & \text{if } t_s > t' \end{cases} \quad (4)$$

The reward structure draws inspiration from PRIMAL (Sartoretti et al., 2019), which addresses point-to-point navigation in multi-robot teams and provides certain dense and sparse rewards analogous to our proposed approach. While not explicitly detailed, the references inform penalization of inefficient actions and rewards for exploration. Additionally, our sparse rewards are scaled with high positive values compared to dense rewards, incentivizing agents to achieve complete area coverage and prioritize remaining unvisited cells. As the most closely related work, PRIMAL serves as a reference point, though our framework introduces novel contributions tailored to multi-agent area coverage challenges.

Using the rewards obtained by a single agent, the joint rewards of all agents can be calculated as the summation of rewards from all agents using Eq. (5).

$$R(S, A, S') = \sum_{i=1}^N R_i(S, A, S') \quad (5)$$

Using the given reward assignment, a policy can be trained to enable the selection of actions that help maximize the joint rewards of the agents given the current observations and states. One approach to model the policy of Dec-POMDP scenarios is to use a joint policy, which is a tuple of local policies, one for each agent in the system as discussed by Oliehoek et al. (2016) and Triantafyllou, Singla, and Radanovic (2022). The history of each agent will be represented explicitly to ensure that each agent can access the histories of other agents. The

Action-Observation History (AOH) for agent i , h_i^A , shown in Eq. (6) represents the sequence of actions and observations taken and received respectively by the agent at a particular timestep up until timestep t . The set of possible AOHs of all timesteps for agent i is represented by H_i^A .

$$h_i^A = (a_i^0, o_i^0, \dots, a_i^t, o_i^t) \quad (6)$$

A deterministic policy for an agent i , π_i , becomes a function of the history when mapping the AOH to actions that contain all the information that the agent would have. Using this logic, the value, $V(\pi)$, of the joint policy, π , is the expected cumulative reward across all steps and agents given the actions prescribed by the policy until the horizon is achieved which is shown in Eq. (7). This value serves as a metric for assessing the quality of the policy and the coordinated manner in which the agents collectively choose their actions. The goal would be to develop an optimal policy that can maximize the total expected cumulative reward which starts at an initial distribution across states b_0 .

$$V^\pi(s) = E\left[\sum_{t=0}^{h-1} \gamma^t R(\vec{a}_t, s_t) | s, \pi\right] \quad (7)$$

2.3. Proximal Policy Optimization (PPO) algorithms

To maximize the value function documented in Section 2.2 during the centralized training process for cooperating agents, policy-based methods involving actor-critic architecture are used. The process involves a centralized critic that helps to exchange additional information during the training of decentralized actors. PPO algorithms are one such family of policy gradient methods within reinforcement learning that use the actor-critic architecture (Yu et al., 2022). The algorithms are capable of maintaining training stability to ensure that policy oscillations and divergence are mitigated. The favorable characteristics of sample efficiency and parallelization enable agents to balance between exploration and exploitation while leveraging a shared experience buffer to learn simultaneously during training.

Other common MARL algorithms include the Monotonic Value Function Factorization for Deep Multi-Agent Reinforcement Learning (QMIX) and the Multi-Agent Deep Deterministic Policy Gradient (MADDPG). QMIX is a value-based algorithm that learns decentralized policies for each agent by decomposing the joint action-value function into per-agent value functions, which are combined using a mixing network. However, QMIX is more difficult to scale to more complex and dynamic environments with a larger number of agents or when agent behaviors become more unpredictable.

MADDPG is an off-policy algorithm where each agent has its own critic that conditions on the global state and the actions of all agents but makes decisions based only on its own observations. This algorithm is effective in environments with continuous action spaces and varying degrees of cooperation and competition. However, due to the off-policy nature of MADDPG, this would require more careful parameter tuning which will cause performance instability.

In contrast, PPO is suitable for this study on complete area coverage using partial observations because of the on-policy nature of the algorithm in which the policy updates are made based on current environmental interactions. This allows the robots to adapt reliably to changing environmental conditions. Additionally, PPO's entropy regularization also promotes the exploration of new areas while still exploiting the knowledge gained from the covered regions, which helps the robots to learn to balance between exploration and exploitation. Lastly, PPO can handle multiple interacting agents to allow them to learn policies that ensure that each robot's movements cover different parts of the area while still maintaining available space for other robots.

PPO algorithms utilize multiple epochs of stochastic gradient ascent to perform each policy update as proposed by Schulman, Wolski,

Dhariwal, Radford, and Klimov (2017). The major component of PPO revolves around the Clipped Surrogate Objective Function, $J^{CLIP}(\theta)$, designed to improve training stability and limit the change of policy at each step. The Clipped Surrogate Objective Function of PPO is shown in Eq. (8):

$$J^{CLIP}(\theta) = \hat{E}_t[\min(r_t(\theta)\hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)\hat{A}_t)] \quad (8)$$

Within the objective function, the ratio function shown in Eq. (9), $r_t(\theta)$, represents the probability of taking action a_t at state s_t in the current policy divided by the previous one. Given that π_θ is the action probability using the current policy and $\pi_{\theta_{old}}$ is the action probability of the old policy, a ratio function more than 1 indicates that action a_t at state s_t is more likely in the current policy than the old one.

$$r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta_{old}}(a_t|s_t)} \quad (9)$$

The unclipped component, $r_t(\theta)\hat{A}_t$, multiplies the probability ratios by the advantage while the clipped component, $\text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon)$, constrains the objective function within the range $[1 - \epsilon, 1 + \epsilon]$ by penalizing changes that lead to a ratio away from 1. The clipping ensures that the policy update by the current policy is not too large from the previous policy which could lead to divergence. The final objective will be the selection of the minimum of the clipped and unclipped components based on the ratio or advantage situation.

3. Training and simulation results

This section documents the training and validation results of the proposed framework for complete area coverage by multiple reconfigurable robots. The training is implemented on different 2D maps that resemble indoor spatial areas shown in Fig. 5. The training maps are split into 4 different datasets to cater for training using a different number of agents to showcase the framework's ability to scale. The first dataset (Dataset 1) is a basic dataset, containing examples of training maps displayed in Figs. 5(a) to 5(d), for training in sparsely dense environments using 2 agents. The second dataset (Dataset 2) is enhanced, containing training map examples displayed in Figs. 5(a) to 5(l), to cater for more obstacle-dense and larger environments using 2 agents. The third (Dataset 3) and fourth (Dataset 4) datasets, containing training map examples displayed in Figs. 5(m) to 5(p) and Figs. 5(q) to 5(t) respectively, are used to train 3 and 4 agents correspondingly.

The maps were designed to include common obstacles that resulted in narrow spaces and tight entry points. This is to simulate areas representative of application scenarios such as food centers, warehouses, or office areas. This requires the robots to leverage the shape-changing reconfiguration capabilities to not only access and pass through these areas but also achieve complete area coverage. The black rectangular cells indicate the obstacles and the orange cells represent tight spaces with a bonus reward discussed in Section 2.2, while the white cells are free areas that the robots can occupy. To simulate real-world deployments and prevent bias, the agents have randomized initial starting positions, and the maps are continuously toggled during training to prevent overfitting.

The environment was created using OpenAI's PettingZoo library (Terry et al., 2021), which is the multi-agent version of the Gymnasium library, that provides an Application Programming Interface (API) to model the spaces and actions while enabling the agents to communicate between the environment and algorithms. From the library, there are 4 main functions:

- Initialization: The environment is created with the major starting parameters with the action and observation spaces.
- Reset function: Resets the robots' initial starting positions, rewards, and training map.

Table 1

Hyperparameters of the PPO algorithm used for training.

Hyperparameter	Value
Learning rate	5×10^{-5}
Training batch size	4096
Mini batch size	128
Lambda	1
Number of SGD iterations	30
Value loss coefficient	1
Entropy loss coefficient	0
Clip parameter	0.2
Number of workers	10

- Step function: Updates the states and observations of each robot while allocating the respective rewards based on the action taken. The joint rewards are calculated once all robots have completed their actions. The termination and truncation functions identify if the episode should end based on complete coverage or maximum timesteps (indicating a recursive loop) achieved respectively.
- Render: Visualizes the trajectories of the robots within the environment.

To train this environment, Ray Rllib (Liang et al., 2017) was utilized which is an open-source library with contemporary reinforcement learning algorithms that have been effective when implemented on a multitude of tasks. The library enables optimized and scalable implementations of the algorithms on custom environments and supports the Tensorflow framework which is used in this paper. The training was conducted on Ubuntu 20.04 using NVIDIA GeForce RTX 3080 GPU and Intel Core i7 CPU. The hyperparameters of the PPO algorithm used for training are summarized in Table 1. The hyperparameters were selected based on optimized values proposed by Schulman et al. (2017), Yu et al. (2021) for cooperative multi-agent settings. The number of Stochastic Gradient Descents (SGD) provides the number of iterations or updates performed using the SGD algorithm to compute the gradient of a loss function with respect to the model parameters using a mini-batch of data. The number of workers indicates the number of parallel agents or environmental runners used to collect experiences simultaneously from the environment during training.

3.1. Analysis of simulation results

From the training process on the basic dataset (Dataset 1) using PPO, the mean reward graph displayed in Fig. 6 demonstrates an initial dip to a minimum mean reward of -1.08×10^4 at 8.68×10^5 environmental steps. However, a consistently sharp increase is achieved afterwards by the policy where the mean reward plateaus and converges at a value of 3095 at 2.00×10^6 environmental steps that required 7.5 h. A maximum convergence mean reward value of 3317 is achieved at 2.67×10^6 environmental steps. The trend indicates that the policy is able to learn meaningful representations of the agents interaction with the environment and make better decisions. The smooth learning curve and convergence also demonstrates that the behavior of the policy is becoming stable to achieve successful learning and adaptation for the agents.

To cross-check the convergence results, the mean episode length graph displayed in Fig. 7 displays an initial sharp increase to reach a maximum mean episode length of 3534 at 8.68×10^5 environmental steps which correspond to the lowest mean reward achieved. The mean episode length decreases sharply and achieves convergence at a mean episode length of 178 at 2.00×10^6 . A minimum mean episode length of 128 is achieved at 2.67×10^6 environmental steps which is consistent with the value of the environmental steps when the maximum mean reward was attained.

Analyzing the remaining datasets, Datasets 2, 3, and 4 each achieved a convergence mean reward value of 2732, 4152, and 5526 at 2.55×10^6 ,

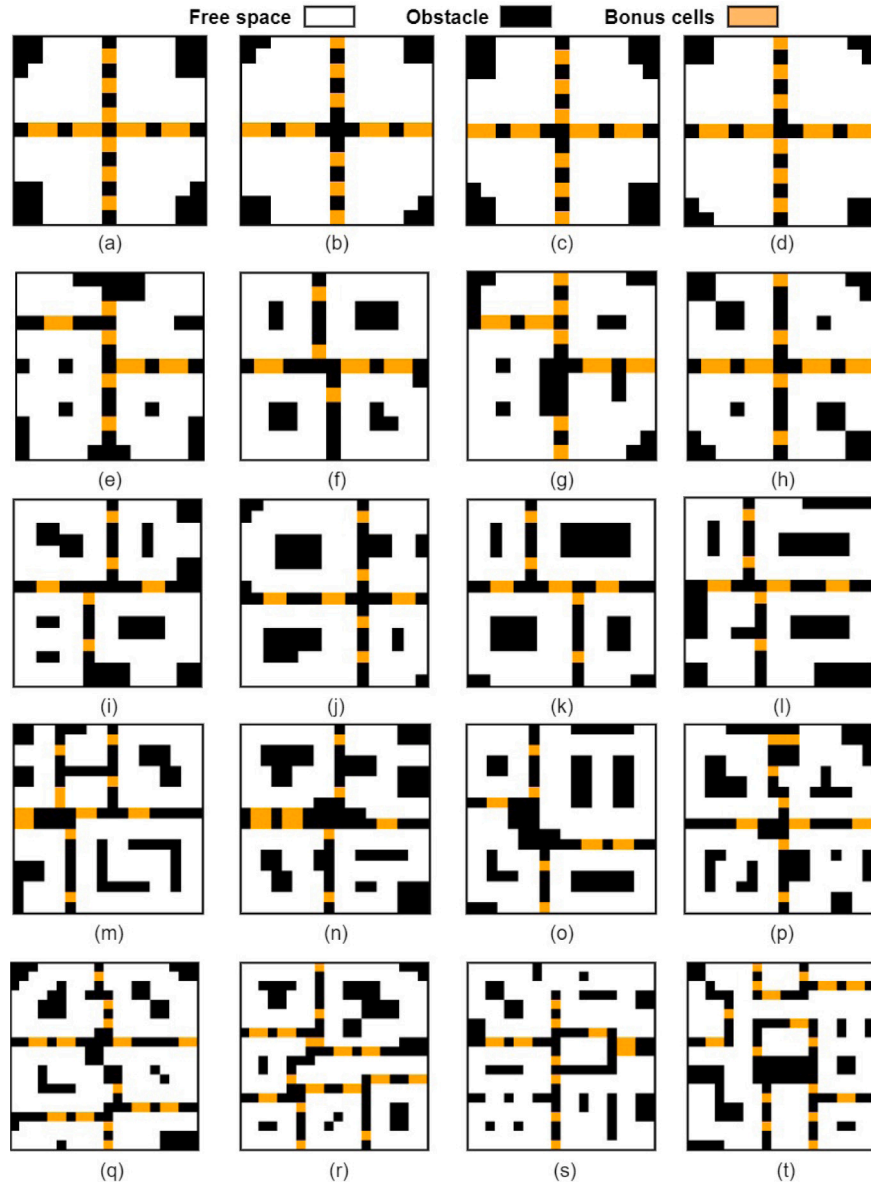


Fig. 5. Visualization of training map examples used in the curation of the 4 datasets for MARL training. (a) to (d): Basic dataset for 2 agents. (a) to (l): Enhanced dataset for 2 agents. (m) to (p): Dataset for 3 agents. (q) to (t): Dataset for 4 agents. Bonus cells indicate areas of tight spaces.

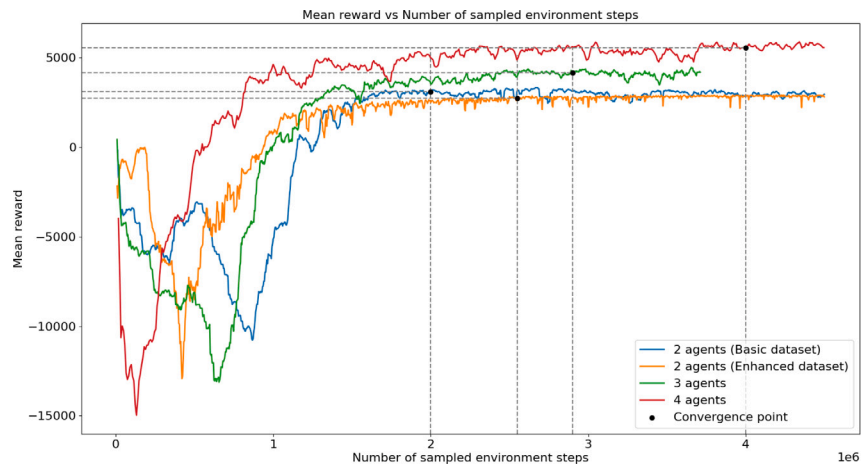


Fig. 6. Mean reward graph of the training processes.

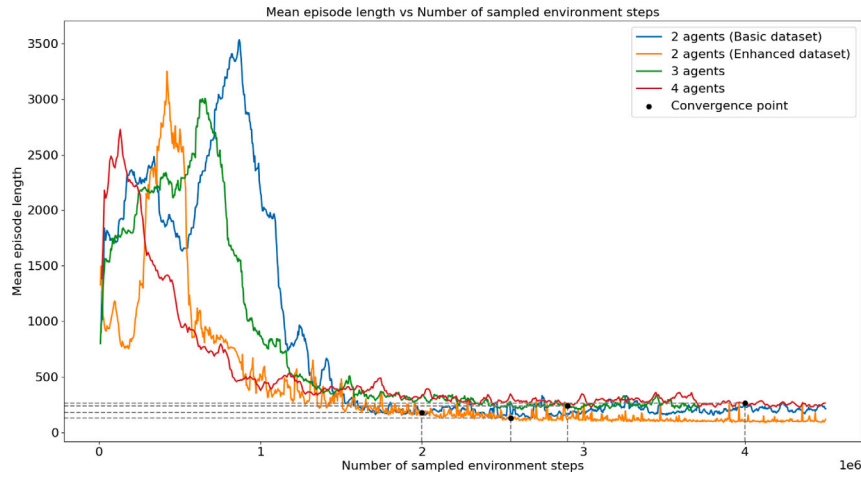


Fig. 7. Mean episode length graph of the training processes.

2.90×10^6 , and 4.00×10^6 environmental steps respectively. At the respective convergence timesteps for each dataset, a convergence mean episode length of 128, 238, and 266 were achieved correspondingly. The training duration to achieve convergence was 11.2 h, 9.7 h, and 8 h for Datasets 2, 3, and 4 respectively. The longer convergence time for Dataset 2 could be attributed to the increased obstacle density in the maps of the dataset. Thus, the policy required a longer time to learn the appropriate joint actions for the robots to achieve complete coverage. Additionally, policy training using Dataset 4, which uses 4 agents, achieved a faster convergence time as compared to Dataset 3, which uses 3 agents. The difference could be due to the stochastic nature of the policy learning process, in which the policy might be able to learn and coordinate the joint actions of the robots better implicitly in Dataset 4 to achieve complete area coverage faster as compared to Dataset 3.

To further verify the training process, a maximum mean reward value of 2874, 4348, and 5848 was achieved correspondingly for Datasets 2, 3, and 4 at the respective timesteps of 4.03×10^6 , 2.62×10^6 , and 3.80×10^6 . This environmental step also corresponds consistently to the minimum mean episode lengths of 92, 207, and 228 achieved by each of the respective datasets.

The smooth learning curves obtained using the PPO algorithm showcase the potential and ability of the algorithm to provide reliable policy updates for multi-agent training, even when scaled up to a greater number of agents. The algorithm helps to also minimize fluctuations that could potentially destabilize the policy's learning process.

3.2. Policy validation

To assess the effectiveness the proposed framework for complete area coverage, policy validation was conducted to evaluate the policy's generalization capabilities. The process involved using the saved weights of the policy that achieved the highest mean reward and lowest episode length. For each dataset, the weights were evaluated on new and unseen test maps (previously unused during the training procedure). The spatial layouts of the test maps are displayed in Fig. 8 for 2 agents and in Fig. 9 for 3 and 4 agents. These test maps included different varying obstacle positions and densities.

From the validation experiment, Table 2 documents the number of timesteps taken by the robots to achieve complete area coverage for each map. From the results, Test Maps 1, 2, and 3 required 87, 76, and 90 timesteps respectively to cover all areas. These values are approximately half of the mean episode length of 178 and less than the minimum episode length of 128 obtained during the training process for Dataset 1 as discussed in Section 3.1. Similarly, Test Maps 4, 5, and 6, required 85, 73, and 88 timesteps respectively which were lower

Table 2

Total episode length and Reconfiguration Steps (RS) required to achieve complete area coverage using the trained policies for the respective test maps.

Test map number	Dataset	Episode length	RS	Area covered (%)
1	1	87	6	100 (see Fig. 8(a))
2	1	76	8	100 (see Fig. 8(b))
3	1	90	6	100 (see Fig. 8(c))
4	2	85	8	100 (see Fig. 8(d))
5	2	73	5	100 (see Fig. 8(e))
6	2	88	6	100 (see Fig. 8(f))
7	2	83	7	100 (see Fig. 8(g))
8	2	105	10	100 (see Fig. 8(h))
9	2	94	8	100 (see Fig. 8(i))
10	3	203	10	100 (see Fig. 9(a))
11	3	215	18	100 (see Fig. 9(b))
12	3	232	21	100 (see Fig. 9(c))
13	4	234	17	100 (see Fig. 9(d))
14	4	258	22	100 (see Fig. 9(e))
15	4	226	15	100 (see Fig. 9(f))

than the minimum episode length of 92 for Dataset 2, even in an obstacle-dense environment. Even when the test environment size was increased, Test Maps 7, 8, and 9, still achieved complete coverage in 83, 105, and 94 steps respectively which is lower than the average episode length of 128 for Dataset 2. Similarly, the Test Maps 10, 11, 12, 13, 14, and 15 achieved the corresponding episode lengths of 203, 215, 232, 234, 258, and 226 which were still below the mean episode lengths of 238 and 266 for Dataset 3 and 4 respectively. The results indicate the ability of the trained policy to perform and generalize well even when implemented in different unseen environments.

Figs. 8 and 9 display the visualization of the coordinated waypoint trajectories taken by each of the robots in the test maps. In all the test maps, it can be observed that each robot attempts to cover areas that are unvisited while also avoiding previously covered areas and trajectories by the other robot. For example, in Test Map 1, the first (Smorphi1) and second (Smorphi2) robots each extensively cover the top right and bottom left sections of the map respectively. Both robots also partially cover the top left and bottom right areas of the map to ensure that any remaining scattered areas uncovered by the other robot are visited. Similarly, in Test Map 3, Smorphi1 (initialized at the top left section of the map) takes an anticlockwise route while Smorphi2 (initialized at the bottom right section of the map) utilizes a clockwise route to each extensively cover the bottom left and top right sections of the map respectively. In Test Map 2, Smorphi1 and Smorphi2 each extensively cover the left and right segments of the map respectively. This trend of coordinated coverage planning is also observed among the robots in denser environments within Test Maps 4 to 6 or even when

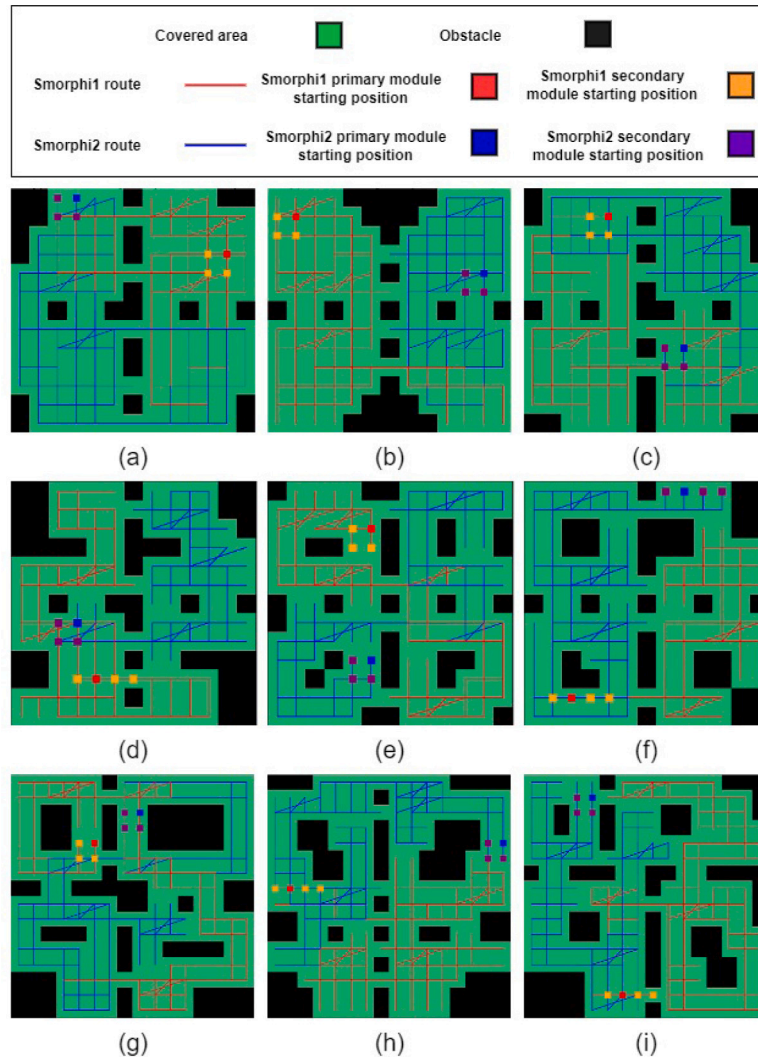


Fig. 8. Waypoint trajectories of the robots to achieve complete area coverage for 2 robots.

the environment size and number of participating robots are increased in Test Maps 7 to 15. The selection of trajectories could potentially be attributed to the communication of the robots' current positions between each other in order to facilitate in complete area coverage.

Additionally, the total reconfiguration steps performed by the agents are summarized in Table 2. Specific locations of the performed reconfiguration steps in each test map are visually represented by the crossed trajectory lines in Figs. 8 and 9. In all maps and across the agents, the reconfiguration steps are primarily conducted in areas near tight and narrow spaces. This indicates the robots' ability to initiate reconfiguration strategically at these specific areas in order to transform into the most suitable shape for navigating through the respective tight spaces and accessing another segment of the map.

3.3. Baseline comparisons

To further assess the effectiveness of the proposed MARL approach, the area coverage percentage is compared with baselines comprising of using different robot combinations. The baselines involve using multiple fixed I and O shaped morphology robots and calculating the percentage of area coverage on Test Maps 1 to 3 evaluated in Section 3.2. A total of 5 baseline combinations are used where 2 combinations utilize fixed I shaped robots whose starting positions are initialized on the same and opposite halves of the environment. Another 2 combinations include fixed O shaped robots also initialized on the

same and opposite halves while the last combination consists of using one I and one O shaped robot.

Table 3 collates the maximum coverage area achieved by each combination across the 3 test maps while Fig. 10 displays the spatial visualization of the covered and accessible areas of Test Maps 1, 2, and 3 respectively. From the results, it can be analyzed that none of the 5 baselines have the capacity to achieve complete area coverage due to the lack of reconfiguration and the presence of multiple tight spaces. From Figs. 10a, 10b, 10c, 10g, 10h, and 10i, when the robots are initialized on the same half or side of the environment, the maximum area covered ranges from 46.21% to 48.97% only. This is because lack of reconfiguration capabilities disallows the robots to navigate pass the obstacles and tight spaces that separate the two halves of the environment. Thus, the robots have totally no access to the other half of the map which would make deployment ineffective. From Figs. 10d, 10e, 10f, 10j, 10k, and 10l, initializing the robots at the opposite sides of the environment does increase the maximum area covered to span from 94.48% to 96.55%. However, complete area coverage is still not achieved due to the inability of accessing the tight spaces arising from the robots' fixed morphologies. Similarly, using a combination of both I and O shaped robots, displayed in Figs. 10m, 10n, and 10o achieves a maximum area coverage ranging from 71.72% to 73.10% as each robot is only capable of clearing certain types of obstacles and tight spaces. In contrast, using the proposed MARL framework on reconfigurable robots, complete area coverage is achieved across the

Table 3

Comparison of maximum area coverage percentage achieved by the proposed framework (highlighted in green) with other robot combinations on the test maps.

Scenario number	Robot shapes	Robot type	Initialization side	Test map	Maximum area covered (%)
1	I	Fixed	Same	1	48.97 (see Fig. 10(a))
2	I	Fixed	Same	2	47.59 (see Fig. 10(b))
3	I	Fixed	Same	3	46.21 (see Fig. 10(c))
4	I	Fixed	Opposite	1	94.48 (see Fig. 10(d))
5	I	Fixed	Opposite	2	94.48 (see Fig. 10(e))
6	I	Fixed	Opposite	3	94.48 (see Fig. 10(f))
7	O	Fixed	Same	1	46.90 (see Fig. 10(g))
8	O	Fixed	Same	2	48.97 (see Fig. 10(h))
9	O	Fixed	Same	3	46.90 (see Fig. 10(i))
10	O	Fixed	Opposite	1	96.55 (see Fig. 10(j))
11	O	Fixed	Opposite	2	96.55 (see Fig. 10(k))
12	O	Fixed	Opposite	3	96.55 (see Fig. 10(l))
13	I, O	Fixed	–	1	72.41 (see Fig. 10(m))
14	I, O	Fixed	–	2	73.10 (see Fig. 10(n))
15	I, O	Fixed	–	3	71.72 (see Fig. 10(o))
16	I, O	Reconfigurable	–	1	100 (see Fig. 1(a))
17	I, O	Reconfigurable	–	2	100 (see Fig. 1(b))
18	I, O	Reconfigurable	–	3	100 (see Fig. 1(c))

test maps irregardless of the robots' starting positions.

The results from both the policy validation and baseline comparisons have showcased the potential effectiveness and usefulness of adopting such a MARL framework using Dec-POMDP to achieve complete area coverage for multiple reconfigurable robots. The framework is a promising proposition for implementation in real-world applications comprising multiple collaborative reconfigurable robots.

3.4. Limitations

Beyond the promising results of the proposed work, there are also limitations involved in the approach. The current framework does not explicitly account for perception uncertainties such as sensor noises or errors. Other forms of real-world uncertainties also include potential latency complications leading to communication delays or localization errors which may result in accumulated positional offsets. However, these issues could be mitigated in the future through sensor fusion or multi-modal sensing to ensure that the errors are minimized across different domains and do not heavily deviate the outcomes of the proposed approach.

The proposed framework is also applied and validated currently for shape-changing reconfigurable robots. Additionally, other classes of reconfigurable robots may also have different reconfiguration durations following the respective hardware specifications. Nevertheless, the approach can still be generalized across other classes of reconfigurable robots, such as size-changing intra-reconfigurable robots. This can be achieved by modifying the appropriate states and actions while reassigning proportional rewards to further prioritize shortened deployment times or strategic reconfigurations by conducting retraining of the policies. However, essential aspects of the proposed framework, such as positional coordinate sharing, observation space parameters and state updates, are still retained. Furthermore, the framework can also be expanded to train on larger environments and extended towards other interesting applications such as search-and-rescue operations, agricultural monitoring, or construction and infrastructure inspection.

Another consideration is that the training process may be computationally expensive due to the modeling of different states in reconfigurable robots. However, the issue is only confined to the training phase of the framework as the computational cost required to deploy the trained policy in test environments is low and can be easily implemented across different robot platforms.

Furthermore, the current reward structure may not fully capture the complexities of the task problem even though effective area coverage is achieved. Future efforts will explore curriculum learning, where

agents begin with simple rewards for basic tasks that evolve into more complex representations as tasks become more challenging. Moreover, integrating physics-based principles, such as robot kinematics and dynamics, into reward engineering could provide a more realistic and effective framework for learning policies, ultimately improving the agents' performances in real-world settings.

Lastly, while the generated training and test maps simulate diverse scenarios to promote environmental generalization, there are still discrepancies between the obstacle density and layout between training and test map configurations. To address this, future work will focus on standardizing the map selection process for training and testing by aligning obstacle density and layout distributions within a quantifiable threshold while retaining controlled randomness. This approach will allow more consistent and interpretable performance comparisons while maintaining the agents' ability to generalize complete area coverage across different environments.

Despite the above mentioned limitations, the proposed framework is a valuable contribution to the domain of reconfigurable robots as there is currently limited research work that focuses on frameworks for reconfigurable robots in multi-agent settings. The contribution using multiple reconfigurable robots to achieve complete area coverage can be utilized to enhance the coordination and efficiency in a multitude of practical tasks.

4. Conclusions

This paper proposed a novel MARL framework that leveraged on the Dec-POMDP to enable multiple shape-changing reconfigurable robots to achieve complete area coverage. The usage of the reconfigurable robot, Smorphi, provided the capacity to navigate through different types narrow spaces by leveraging the robot's shape-changing capabilities. The framework enabled the coordination and learning among multiple reconfigurable robots to achieve complete area coverage through interactions with the environment. The framework also considered the practical limitations of partial observability, the changing state dynamics of the environment and the dependencies of the robots' learning policies with one another.

Through the training process, a joint policy was generated to enable agents to share information so that the set of actions selected by the agents as a whole would maximize the value of the joint policy. From the obtained training results, the utilization of the PPO algorithm achieved a consistent increase and convergence of the mean rewards. This indicates the ability of the algorithm to effectively train a stabilized joint policy to learn meaningful representations from the robots'

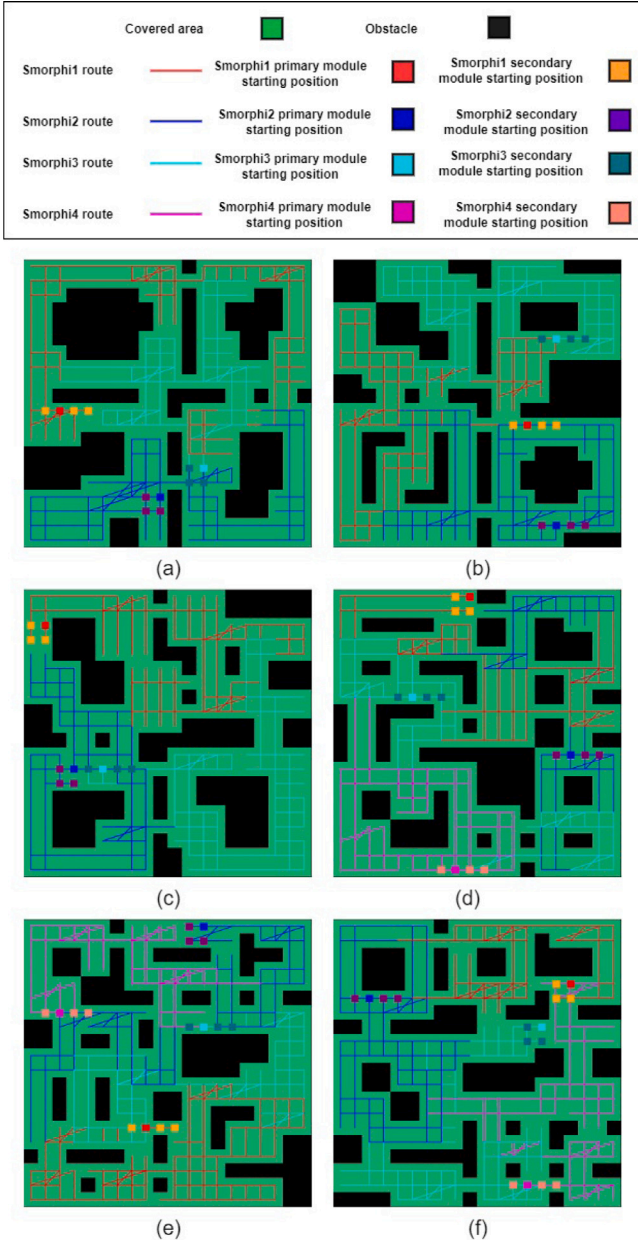


Fig. 9. Waypoint trajectories of the robots to achieve complete area coverage for 3 and 4 robots.

interactions with the environment and one another at each timestep.

To evaluate the generalization capabilities of the framework and policy, the weights of the policy at the timestep with the highest reward achieved during the training process was validated on 3 new test maps with different obstacle placements. In all scenarios, the robots achieved complete area coverage in timesteps lower than the mean and minimum episode length achieved during training, irregardless of starting points. Additionally, the area coverage was compared with baselines using different combinations of multiple fixed morphology robots in which none of the combinations managed to achieve complete area coverage. The policy validation and baseline comparisons have proven the effectiveness of the implemented framework's generalization abilities and also superior performance when compared to multiple fixed morphology robots. The future work of this research has potential to be expanded into different scopes that will focus on:

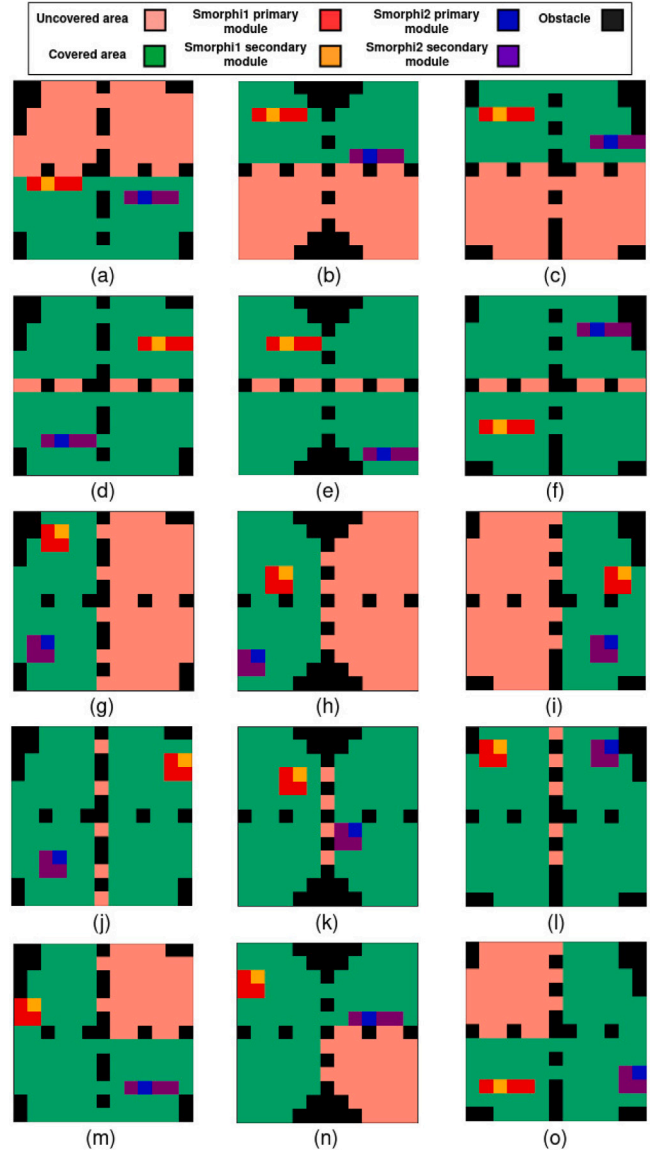


Fig. 10. Visualization of maximum area coverage using different fixed morphology robot combinations.

1. Scaling the framework up to accommodate larger environments, increased number of robots, and also involving different types of reconfigurable robots to form a heterogeneous multi-agent setting. Large environments can be segmented into smaller regions in which the proposed framework can be deployed for each region across a small group of robots. This helps to achieve complete coverage while maintaining computational costs.
2. Training the framework to consider dynamic and unpredictable obstacles within unstructured environments by leveraging on Graph Neural Networks (GNN).
3. Integrating the framework onto actual reconfigurable robots to test out the effectiveness in real-world scenarios.
4. Reducing or maintaining the computational complexity of the Multi-agent Reinforcement Learning training process even when a larger training dataset is used.
5. Leveraging the partial observations of each robot to perceive different environmental landmarks to improve localization for multiple reconfigurable robots.

CRediT authorship contribution statement

J.J.J. Pey: Conceptualization, Methodology, Software, Visualization, Formal analysis, Data curation, Investigation, Writing – original draft. **S.M. Bhagya P. Samarakoon:** Conceptualization, Formal analysis, Validation, Methodology, Supervision, Writing – review & editing. **M.A. Viraj J. Muthugala:** Conceptualization, Validation, Methodology, Supervision Writing – review & editing. **Mohan Rajesh Elara:** Conceptualization, Validation, Supervision, Funding acquisition Project administration, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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